

Class 9: RANBooster

Democratizing Advanced Cellular Connectivity through
Fronthaul Middleboxes

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RANBooster: Democratizing Advanced Cellular Connectivity through Fronthaul Middleboxes

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Microsoft Research · UC Berkeley · ICSI

ACM SIGCOMM 2025, Coimbra

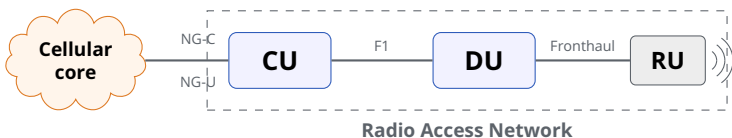
Put a **middlebox** on the RAN fronthaul, and you can add features the base station vendor never shipped.

Paper: dl.acm.org/doi/10.1145/3718958.3754349 Code: github.com/microsoft/RANBooster Talk: youtu.be/wutVaZlgTLs

Recent trends in cellular RAN technologies

5G RAN characteristics:

- Standardized and open interfaces
- Softwarized on commodity hardware (x86, ARM, GPUs)

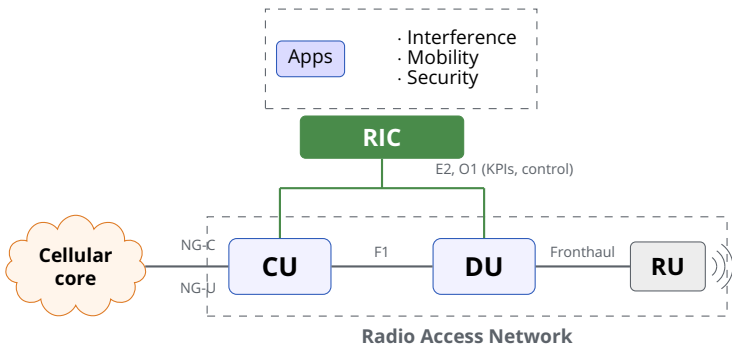


CU / DU / RU = Centralized / Distributed / Radio Unit RIC = RAN Intelligent Controller

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Opportunities and challenges for new players

Disaggregation, virtualization and open interfaces:

- + Opportunity to open the closed cellular ecosystem
- Cannot match the performance of integrated end-to-end solutions

Interference
mitigation?

Mobility
management?

Radio and spectrum
sharing?

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Build RIC apps to enable advanced features?

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Build RIC apps to enable advanced features?

“Open RAN xApps look stillborn”

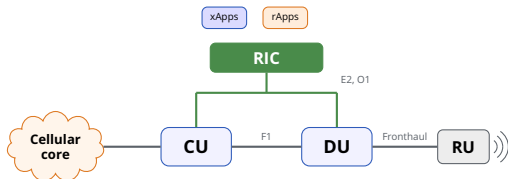
The lack of support from Ericsson and Samsung for the near-real-time RIC has left xApps with nowhere to go.

Iain Morris, Light Reading, December 5, 2024 — the headline the authors put on this slide

How can we help democratize the RAN?

Leverage open interfaces to deploy new features as software middleboxes

- Mirror the evolution of the Internet architecture



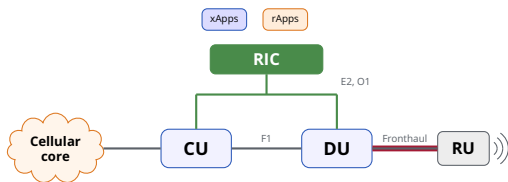
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Our focus: the fronthaul interface

- Supported by all RAN vendors, large and small
- Limited state, simple semantics



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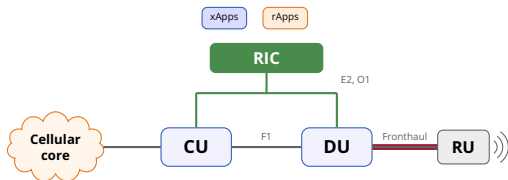
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Common perception: the fronthaul is of limited value

- Low-level data — digitized radio signals (IQ samples)
- Carries latency-sensitive traffic



The first middlebox framework for the RAN

4 reference middleboxes and enterprise-scale deployment

- Distributed Antenna System
- Distributed MIMO
- RU sharing
- Radio resource block monitoring

Left column: built, measured, deployed. Right column: sketched in the paper, not built.

The first middlebox framework for the RAN

4 reference middleboxes and enterprise-scale deployment

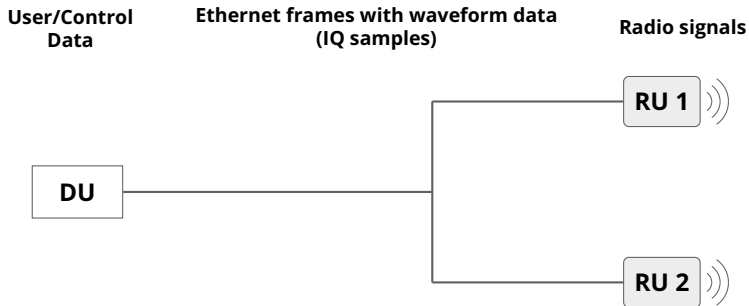
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Other applications

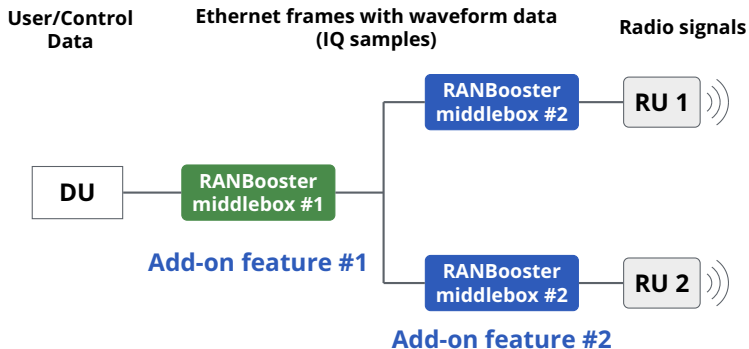
- Sensing
- Security
- Failover and resilience

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High-level RANBooster middlebox operation



High-level RANBooster middlebox operation



The DU still thinks it is talking to its own radios.
Nothing above the fronthaul had to change.

RANBooster packet manipulation primitives

4 supported types of actions

A1: Packet redirection/drop



A2: Packet replication



A3: Packet caching

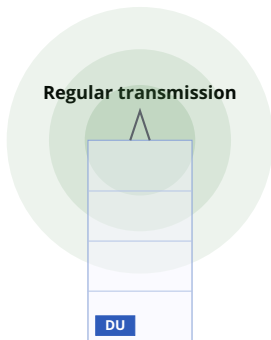


A4: Packet inspection and modification



M = a RANBooster middlebox. Every application in the paper is some combination of A1–A4.

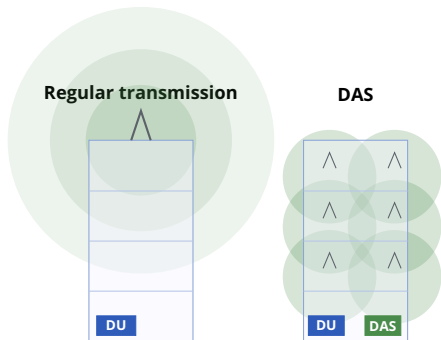
Example 1: Distributed antenna system (DAS)



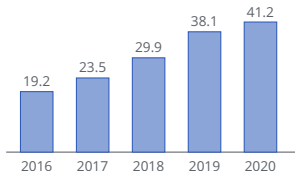
DAS node counts: Boingo Wireless, as cited in the authors' talk

Example 1: Distributed antenna system (DAS)

Replace the high-power antenna with low-power distributed ones



DAS nodes operated by Boingo Wireless worldwide (thousands)

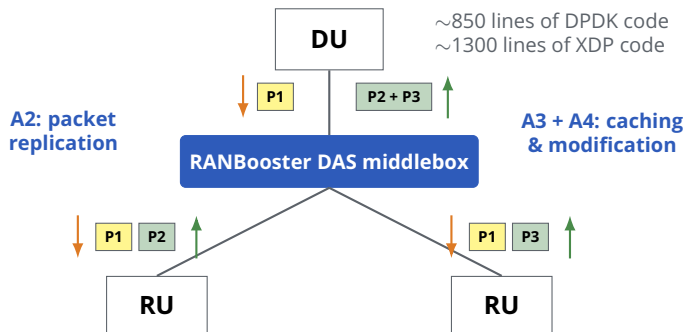


Pros: Better, uniform coverage — standard practice in enterprises, stadiums, airports

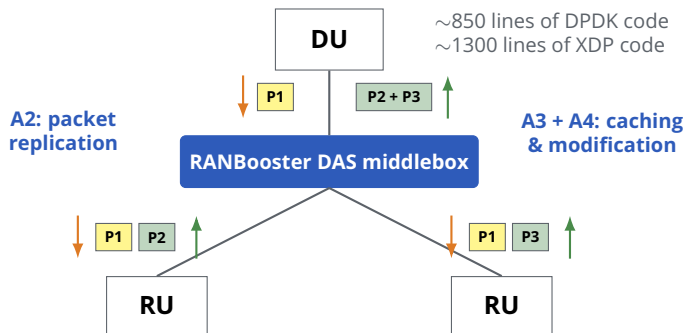
Cons: Very high cost and deployment complexity — hundreds of thousands to millions of dollars

DAS node counts: Boingo Wireless, as cited in the authors' talk

The DAS middlebox: replicate down, combine up



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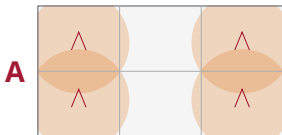
The same binary, with no code changes, ran over the SD-RAN, Capgemini, Radisys and Foxconn stacks — the fronthaul is standard.

RANBooster DAS benefits: cell planning

Microsoft Cambridge 5G testbed — one floor, 4 RUs, 100 MHz of spectrum

How can I do the cell planning?

A: Split 100 MHz into 4×25 MHz cells — no interference, but peak throughput is capped



Downlink throughput across the floor, redrawn schematically from the measured heatmaps in the talk.

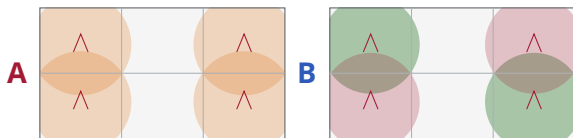
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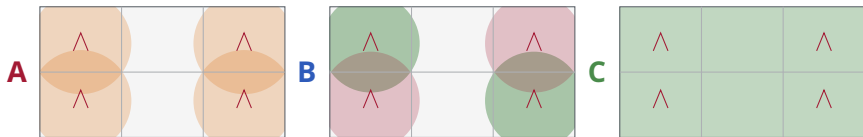
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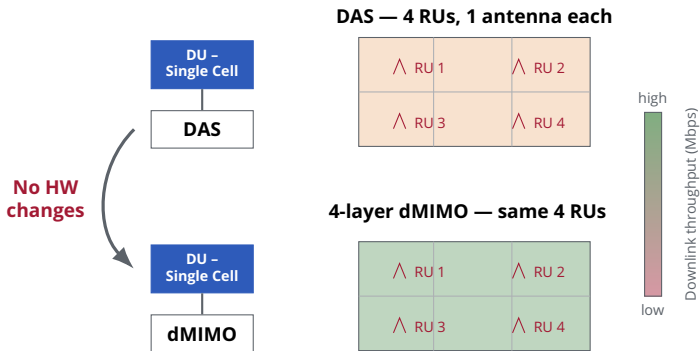
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C: RANBooster DAS: one 100 MHz cell, 4 radios acting as its antennas

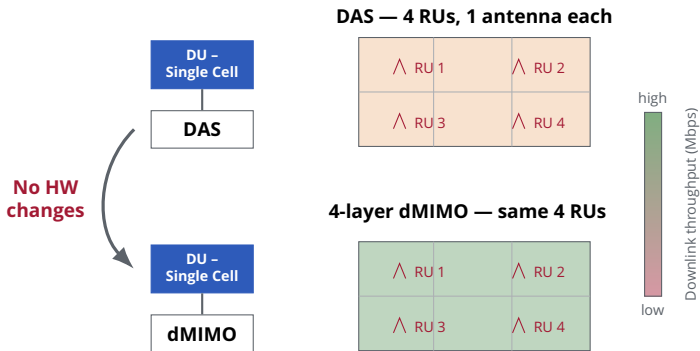


Downlink throughput across the floor, redrawn schematically from the measured heatmaps in the talk.

Example 2: Distributed MIMO (dMIMO)



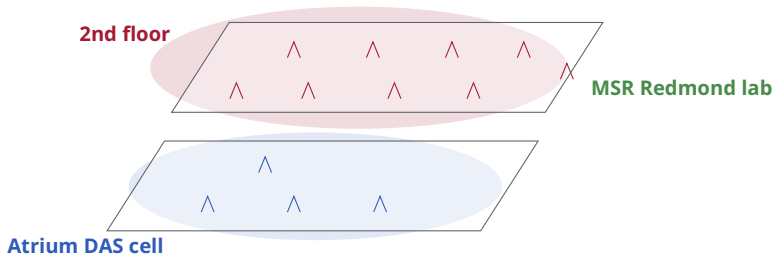
Example 2: Distributed MIMO (dMIMO)



Swapping the DAS middlebox for a dMIMO middlebox more than doubled the capacity of the floor.

RANBooster is already running 24/7 in our deployments

- Cambridge MSR lab: full coverage of 4 floors with 16 RUs
- Redmond MSR lab: full coverage of 2 floors with 13 RUs



Chained middleboxes: one per floor, combined by a building-wide DAS. Measured added latency stayed in the microseconds.

RANBooster: a fronthaul middlebox framework

- Build novel monitoring and control use cases
- Accelerate innovation in the RAN space

Open source reference implementation

`github.com/microsoft/RANBooster`

Four primitives on an interface every vendor already speaks were enough to add features the vendors never shipped.

Q: The DU and the RUs are connected through a fronthaul switch. Where does this software actually run?

A: In software on a server today. Offloading to SmartNICs or programmable switches is the obvious next step — but packet caching (A3) is the action that is not clearly implementable on a switch. Open problem.

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Q: You said a middlebox on a latency-sensitive path might be a problem. Was it?

A: No. The actions mostly rewrite header fields, which costs microseconds. That is what allowed scaling to eight or nine radios, and chaining one middlebox per floor underneath a building-wide DAS.

What this paper is really arguing

The move

- The official extension point (the RIC) went unsupported.
- So use the interface vendors had to implement.
- Middleboxes: the Internet's own answer to a frozen architecture.

Worth arguing about

- Middleboxes made the Internet hard to evolve. Same trap here?
- Who operates a box that sees every user's raw waveform?
- Does this scale past one enterprise floor?

Where else in this course have we seen a closed interface pried open by someone standing on the wire?